*Local null controllability of a parabolic system using a Neumann boundary control**

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Outline of the talk

- **Introduction**
- **Linearized system**
  - Method of moments
  - Global null controllability
- **Linearized system with source term**
  - Source term method
  - Weighted L^2-estimates and null controllability
- **Nonlinear system**
  - Banach fixed point argument
  - Local null controllability

Introduction

- Let $T > 0$. We consider

$$(NL) \quad \begin{cases} \partial_t y - \partial_x^2 y = f(y, z, \int_0^1 y, \int_0^1 z), & \text{in } (0,T)\times(0,1), \\ \partial_t z - \partial_x^2 z = g(y, z, \int_0^1 y, \int_0^1 z), & \text{in } (0,T)\times(0,1), \\ \partial_x y(t,0) = q(t), \quad \partial_x z(t,0) = 0, & \text{for } t \in (0,T), \\ \partial_x y(t,1) = \partial_x z(t,1), & \text{for } t \in (0,T), \\ y(t,1) + z(t,1) + \alpha \partial_x y(t,1) = 0, & \text{for } t \in (0,T), \\ y(0,x) = y_0(x), \quad z(0,x) = z_0(x), & \text{in } (0,1). \end{cases}$$

- $\alpha \geq 0$ is some real parameter
- $q \in L^2(0,T)$ is a control input
- $(y_0, z_0) \in (L^2(0,1))^2$ is a given initial state.

Introduction

- f, g are nonlinear functions given as

$$\begin{cases} f(y, z, \int_0^1 y, \int_0^1 z) = -yz + ay^2 + bz^2 + r_1(t)y, \\ f(y, z, \int_0^1 y, \int_0^1 z) = yz + cy^2 + dz^2 + r_2(t)z. \end{cases}$$

- $a, b, c, d \in L^\infty((0, T) \times (0, 1))$
- $r_j(t) = \alpha_j \int_0^1 (\psi_{1,j}(x)y(t, x) + \psi_{2,j}(x)z(t, x))\, dx, \quad j = 1, 2,$
  where $\alpha_j \in \mathbb{R}$ and $\psi_{1,j}, \psi_{2,j} \in L^\infty(0, 1)$ for $j = 1, 2$.
- This kind of system is commonly known as the "Lotka-Volterra" model with diffusion.
- This model characterize the dynamics of a biological system where two species, prey and predator, interact between each other.
- Our goal is to put an external control force only on one species from the boundary end to locally control both the species at a given time T.

- Let H and U be Hilbert spaces and $T > 0$. We consider the following control system:

$$(PDE) \quad \begin{cases} u'(t) = Au(t) + Bf(t), & t \in (0, T), \\ u(0) = u_0, \end{cases}$$

  - $A : \mathcal{D}(A) \subset H \to H$ is closed and densely defined linear operator.
  - $B : U \to H$ is the control operator (bounded/unbounded).
  - $u : [0, T] \to H$ represents the state vector,
  - $f : [0, T] \to U$ represents the control vector,
  - $u_0 \in H$ is the initial state.

- We also assume that (PDE) has a unique weak solution u in

$$\mathcal{C}^0([0, T]; H).$$

Definition (Controllability)

Let $T > 0$ be given. We say the system (PDE) is

- **exactly controllable** at time T in H if, for given initial state $u_0 \in H$ and final state $u_T \in H$ there exists a control $f \in L^2(0, T; U)$ such that

$$u(T) = u_T.$$

- **approximately controllable** at time T in H if, for given $u_0, u_T \in H$ and given $\epsilon > 0$, there exists a control $f_\epsilon \in L^2(0, T; U)$ such that

$$\|u_\epsilon(T) - u_T\| \leq \epsilon.$$

- **null controllable** at time T in H if, for any given initial state $u_0 \in H$ there exists a control $f \in L^2(0, T; U)$ such that

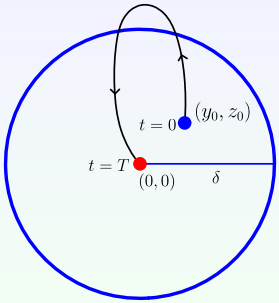
$$u(T) = 0.$$

Small-time local null controllability

Definition

We say the system (NL) is **small-time locally null controllable** around the equilibrium $(0,0)$ in $(L^2(0,1))^2$ if, for any given $T>0$ there is a $\delta>0$ such that for chosen initial state $(y_0,z_0)\in(L^2(0,1))^2$ with $\|(y_0,z_0)\|_{(L^2(0,1))^2}\leq\delta$, there exists a control $q\in L^2(0,T)$ satisfying

$$(y(T),z(T))=(0,0).$$



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- The linearized system around the equilibrium point (0, 0) is given by

$$(LS) \quad \begin{cases} \partial_t y - \partial_x^2 y = 0, & \text{in } (0,T)\times(0,1), \\ \partial_t z - \partial_x^2 z = 0, & \text{in } (0,T)\times(0,1), \\ \partial_x y(t,0) = q(t), \quad \partial_x z(t,0) = 0, & \text{for } t \in (0,T), \\ \partial_x y(t,1) = \partial_x z(t,1), & \text{for } t \in (0,T), \\ y(t,1) + z(t,1) + \alpha \partial_x y(t,1) = 0, & \text{for } t \in (0,T), \\ y(0,x) = y_0(x), \quad z(0,x) = z_0(x), & \text{in } (0,1). \end{cases}$$

- Here $\alpha \geq 0$, $(y_0, z_0) \in (L^2(0,1))^2$ and $q \in L^2(0,T)$.

Step 1: Linearized system

Well-posedness

- We define the operator $(\mathcal{A}, \mathcal{D}(A))$ as

$$\begin{cases} \mathcal{D}(\mathcal{A}) := \{(u,v) \in (H^2(0,1))^2 \ : \ u'(0) = 0, v'(0) = 0, u'(1) = v'(1), \\ \qquad\qquad\qquad\qquad\qquad\qquad u(1) + v(1) + \alpha u'(1) = 0\} \\ \mathcal{A} = \begin{pmatrix} -\partial_x^2 & 0 \\ 0 & -\partial_x^2 \end{pmatrix} \end{cases}$$

- $(\mathcal{A}, \mathcal{D}(A))$ generates an analytic semigroup in the space $(L^2(0,1))^2$.
- The linear system (LS) admits a unique weak solution

$$y, z \in \mathcal{C}^0([0,T]; L^2(0,1)) \cap L^2(0,T; H^1(0,1)).$$

- Moreover, we have the energy estimate

$$\|(y,z)\|_{\mathcal{C}^0([0,T];(L^2(0,1))^2)} + \|(y,z)\|_{L^2(0,T;(H^1(0,1))^2)} \leq Ce^{CT} \left(\|(y_0,z_0)\|_{(L^2(0,1))^2} + \|q\|_{L^2(0,T)} \right)$$

for some constant $C > 0$ independent of T.

Theorem

- *Let $T > 0, (y_0, z_0) \in (L^2(0,1))^2$ and $\alpha \geq 0$ be given. Then there exists a control $q \in L^2(0,T)$ such that the solution (y, z) to (LS) satisfies*

$$(y(T,\cdot), z(T,\cdot)) = (0,0) \text{ in } (0,1).$$

- *Moreover, we have the following estimate*

$$\|q\|_{L^2(0,T)} \leq M e^{\frac{M}{T}} \|(y_0, z_0)\|_{(L^2(0,1))^2}$$

for some constant $M > 0$ independent of T.

- Let $T > 0$ and $\alpha \geq 0$ be given.

Lemma (Equivalent criterion for null controllability)

The system (LS) is null controllable at time T if, and only if, there exists a control $q \in L^2(0,T)$ such that for any $(\zeta_T, \theta_T) \in (L^2(0,1))^2$:

$$\int_0^T q(t) \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \left(e^{-(T-t)\mathcal{A}^*}(\zeta_T, \theta_T)\right)(0) \right\rangle dt = \left\langle (y_0, z_0), e^{-T\mathcal{A}^*}(\zeta_T, \theta_T) \right\rangle.$$

- Replace $(\zeta_T, \theta_T) = \Phi_\lambda$, where Φ_λ is the eigenfunction of $\mathcal{A}^*$ corresponding to the eigenvalue λ.
- We see that the above identity becomes:

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \Phi_\lambda(0) \right\rangle \int_0^T q(t) e^{-(T-t)\lambda} dt = e^{-\lambda T} \langle (y_0, z_0), \Phi_\lambda \rangle.$$

Step 1: Linearized system

Proof of null controllability: Eigenvalues and eigenfunctions of $\mathcal{A}^*$

- $\Lambda^\alpha := \{\lambda_{k,1}, \lambda^\alpha_{k,2} \ : \ k \geq 0\}$ is the set of eigenvalues of $\mathcal{A}^*$, where

$$\lambda_{k,1} = k^2\pi^2, \text{ and } \lambda^\alpha_{k,2} = \begin{cases} (k+\frac{1}{2})^2\pi^2, \text{ if } \alpha = 0, \\ \in (k^2\pi^2, (k+\frac{1}{2})^2\pi^2), \text{ if } \alpha > 0. \end{cases}$$

- For $\alpha > 0$, we have the asymptotic expression of $\lambda^\alpha_{k,2}$:

$$\lambda^\alpha_{k,2} = k^2\pi^2 + \frac{4}{\alpha} + O\left(\frac{1}{k^2}\right) \text{ for } k \text{ large enough.}$$

- The associated eigenfunctions are

$$\Phi_{\lambda_{k,1}}(x) := \begin{pmatrix} \cos(k\pi x) \\ -\cos(k\pi x) \end{pmatrix} \text{ and } \Phi_{\lambda^\alpha_{k,2}}(x) := \begin{pmatrix} \cos(\sqrt{\lambda^\alpha_{k,2}}x) \\ \cos(\sqrt{\lambda^\alpha_{k,2}}x) \end{pmatrix}, \quad k \geq 0.$$

- $\{\Phi_{\lambda_{k,1}}, \ \Phi_{\lambda^\alpha_{k,2}} \ : \ k \geq 0\}$ forms an orthogonal basis in $(L^2(0,1))^2$.

Lemma (Moments problem)

Let $T > 0$ and $\alpha \geq 0$ be given. Then the system (LS) is null controllable at time T if, and only if, there exists a control $q \in L^2(0,T)$ such that

$$(MP) \qquad \int_0^T q(t) e^{-\lambda(T-t)} dt = \frac{e^{-\lambda T}}{\mathcal{B}^* \Phi_\lambda} \langle (y_0, z_0), \Phi_\lambda \rangle \quad \text{for all } \lambda \in \Lambda^\alpha,$$

where $\mathcal{B}^* : (H^1(0,1))^2 \to \mathbb{R}$ is defined as

$$\mathcal{B}^* = \mathbb{1}_{x=0} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Remark

Note that $\mathcal{B}^* \Phi_{\lambda_{k,1}} = 1 \neq 0$ and $\mathcal{B}^* \Phi_{\lambda^\alpha_{k,2}} = 1 \neq 0$ for any $k \geq 0$.

Lemma (Existence of a biorthogonal family (Benabdallah et. al, SIAM J. Control Optim., 2014)**)**

Let $\alpha \geq 0$. Then there exists a family $(p_\lambda)_{\lambda\in\Lambda^\alpha}$ biorthogonal to $(e^{-\lambda(T-\cdot)})_{\lambda\in\Lambda^\alpha}$ in $L^2(0,T)$, that is

$$\int_0^T p_\lambda(t)e^{-\mu(T-t)}dt = \delta_{\lambda,\mu}, \quad \lambda,\mu\in\Lambda^\alpha.$$

Moreover, we have the following estimate

$$\|p_\lambda\|_{L^2(0,T)} \leq Ce^{\frac{C}{T}}e^{\frac{T}{2}\lambda+C\sqrt{\lambda}}, \quad \lambda\in\Lambda^\alpha.$$

- Conditions required:
  - Existence of a biorthogonal family: $\sum_{\lambda\in\Lambda^\alpha}\frac{1}{\lambda} < +\infty$
  - L^2-bounds of the biorthogonal family: $|\lambda-\mu|\geq\rho$, for all $\lambda,\mu\in\Lambda^\alpha$, $\lambda\neq\mu$
  - Explicit bound w.r.t. T:
    $\mathcal{N}_\alpha(r)\leq\kappa_0\sqrt{r}$, and $|\mathcal{N}_\alpha(r)-\mathcal{N}_\alpha(s)|\leq\kappa_0(1+\sqrt{|r-s|})$ for all $r,s>0$,
    where $\mathcal{N}_\alpha(r) := \#\{\lambda\in\Lambda^\alpha \ : \ |\lambda|\leq r\}, \quad r>0.$

- Define the control

$$q(t) = \sum_{\lambda \in \Lambda^\alpha} \frac{e^{-\lambda T}}{\mathcal{B}^* \Phi_\lambda} \langle (y_0, z_0), \Phi_\lambda \rangle \, p_\lambda(t), \quad t \in (0, T).$$

- We have

$$\begin{aligned} \int_0^T q(t) e^{-\lambda(T-t)} dt &= \sum_{\mu \in \Lambda^\alpha} \frac{e^{-\mu T}}{\mathcal{B}^* \Phi_\mu} \langle (y_0, z_0), \Phi_\mu \rangle \int_0^T p_\mu(t) e^{-\lambda(T-t)} dt \\ &= \frac{e^{-\lambda T}}{\mathcal{B}^* \Phi_\lambda} \langle (y_0, z_0), \Phi_\lambda \rangle \end{aligned}$$

- Moreover,

$$\begin{aligned} \|q\|_{L^2(0,T)} &\leq C e^{\frac{C}{T}} \|(y_0, z_0)\|_{(L^2(0,1))^2} \sum_{\lambda \in \Lambda^\alpha} e^{-\lambda T} e^{\frac{T}{2}\lambda + C\sqrt{\lambda}} \\ &\leq C e^{\frac{C}{T}} \|(y_0, z_0)\|_{(L^2(0,1))^2} \sum_{\lambda \in \Lambda^\alpha} e^{-\frac{T}{4}\lambda}. \end{aligned}$$

$$C\sqrt{\lambda} \leq \frac{T}{4}\lambda + \frac{C^2}{T}.$$

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Step 2: Source term method

Introduction

- Let $T > 0$ and $\alpha \geq 0$ be given. We consider the following system:

$$(LSS) \quad \begin{cases} \partial_t y - \partial_x^2 y = \xi, & \text{in } (0,T)\times(0,1), \\ \partial_t z - \partial_x^2 z = \eta, & \text{in } (0,T)\times(0,1), \\ \partial_x y(t,0) = q(t), \quad \partial_x z(t,0) = 0, & \text{for } t \in (0,T), \\ \partial_x y(t,1) = \partial_x z(t,1), & \text{for } t \in (0,T), \\ y(t,1) + z(t,1) + \alpha \partial_x y(t,1) = 0, & \text{for } t \in (0,T), \\ y(0,x) = y_0(x), \quad z(0,x) = z_0(x), & \text{in } (0,1). \end{cases}$$

- $q \in L^2(0,T), (y_0, z_0) \in (L^2(0,1))^2$ and ξ, η belongs to some L^2-weighted space.
- We will apply the source term method introduced by Y. Liu, T. Takahashi and M. Tucsnak (ESAIM Control Optim. Calc. Var., 2013).

Step 2: Source term method

Weight functions

- We choose $\beta > 0$ and $\gamma > 1$ such that

$$1 < \gamma < \sqrt{2}, \text{ and } \beta > \frac{\gamma^2}{2-\gamma^2}.$$

- Define the weight functions

$$\begin{cases} \rho_0(t) := e^{-\frac{\beta M}{(\gamma-1)(T-t)}}, \\ \rho_S(t) := e^{-\frac{(1+\beta)\gamma^2 M}{(\gamma-1)(T-t)}} \end{cases} \quad \text{for all } t \in \left[T\left(1-\frac{1}{\gamma^2}\right), T\right]$$

and $\rho_0(t) = \rho_0(T(1-\frac{1}{\gamma^2}))$, $\rho_S(t) = \rho_S(T(1-\frac{1}{\gamma^2}))$ for all $t \in [0, T(1-\frac{1}{\gamma^2})]$.

- ρ_0, ρ_S are continuous non-increasing function on $[0, T]$ such that

$$\rho_0(T) = \rho_S(T) = 0.$$

- Due to the choices of β and γ, we have

$$\frac{\rho_0^2(t)}{\rho_S(t)} \leq 1 \text{ for all } t \in [0, T].$$

Proposition

Let $\alpha \geq 0$ be given. Then, for given any initial state $(y_0, z_0) \in (L^2(0,1))^2$ and source term $(\xi, \eta) \in L^2(0,T;(L^2(0,1))^2)$, the system (LSS) admits a unique weak solution (y, z) satisfying

$$\left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{C^0([0,T];(L^2(0,1))^2)} + \left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{L^2(0,T;(H^1(0,1))^2)} + \left\| \frac{q}{\rho_0} \right\|_{L^2(0,T)}$$
$$\leq C e^{CT + \frac{C}{T}} \left(\|(y_0, z_0)\|_{(L^2(0,1))^2} + \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(0,T;(L^2(0,1))^2)} \right)$$

for some constant $C > 0$ independent of T.

Remark

Note that $(y(T), z(T)) = \rho_0(T) \left(\frac{y(T)}{\rho_0(T)}, \frac{z(T)}{\rho_0(T)} \right) = 0$ in $(0,1)$.

Step 2: Source term method

Proof of null controllability: Systems on small time interval

- Write the interval $(0, T)$ as

$$(0, T) = \cup_{k\geq 0}(T_k, T_{k+1}) \text{ with } T_k := T - \frac{T}{\gamma^k},\ k \geq 0.$$

- Define the sequence $(a_k)_{k\geq 0}$ as

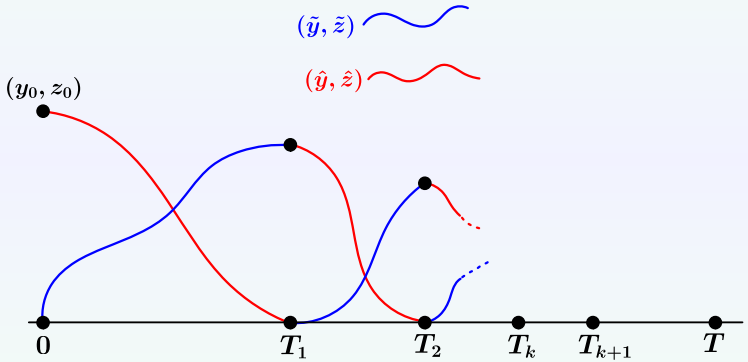
$$a_0 := (y_0, z_0) \text{ and } a_{k+1} := (\tilde{y}(T_{k+1}^-), \tilde{z}(T_{k+1}^-)),\ \text{for all } k \geq 0.$$

- Consider the systems in $(T_k, T_{k+1}) \times (0, 1)$:

$$\begin{cases} \partial_t \tilde{y} - \partial_x^2 \tilde{y} = \xi, \\ \partial_t \tilde{z} - \partial_x^2 \tilde{z} = \eta, \\ \partial_x \tilde{y}(t,0) = 0, \quad \partial_x \tilde{z}(t,0) = 0, \\ \partial_x \tilde{y}(t,1) = \partial_x \tilde{z}(t,1), \\ \tilde{y}(t,1) + \tilde{z}(t,1) + \alpha \partial_x \tilde{y}(t,1) = 0, \\ \tilde{y}(T_k^+, x) = 0, \quad \tilde{z}(T_k^+, x) = 0. \end{cases} + \begin{cases} \partial_t \hat{y} - \partial_x^2 \hat{y} = 0, \\ \partial_t \hat{z} - \partial_x^2 \hat{z} = 0, \\ \partial_x \hat{y}(t,0) = \hat{q}_k(t), \quad \partial_x \hat{z}(t,0) = 0, \\ \partial_x \hat{y}(t,1) = \partial_x \hat{z}(t,1), \\ \hat{y}(t,1) + \hat{z}(t,1) + \alpha \partial_x \hat{y}(t,1) = 0, \\ (\hat{y}(T_k^+, \cdot), \hat{z}(T_k^+, \cdot)) = a_k. \end{cases}$$

Step 2: Source term method

Proof of null controllability: Systems on small time interval



Step 2: Source term method

Proof of null controllability: Estimate of the control

- Energy estimate:

$$\|a_{k+1}\|_{(L^2(0,1))^2} \leq Ce^{CT} \|(\xi,\eta)\|_{L^2(T_k,T_{k+1};(L^2(0,1))^2)}, \quad \text{for all } k \geq 0.$$

- Null controllability implies existence of a control $\hat{q}_k \in L^2(T_k, T_{k+1})$ satisfying the estimate

$$\|\hat{q}_k\|_{L^2(T_k,T_{k+1})} \leq Me^{\frac{M}{T_{k+1}-T_k}} \|a_k\|_{(L^2(0,1))^2}, \quad \text{for all } k \geq 0$$

such that $(\hat{y}(T_{k+1}^-), \hat{z}(T_{k+1}^-)) = (0,0)$.

- This implies

$$\begin{aligned}
\|\hat{q}_{k+1}\|_{L^2(T_{k+1},T_{k+2})} &\leq Me^{\frac{M}{T_{k+2}-T_{k+1}}} \|(\xi,\eta)\|_{L^2(T_k,T_{k+1};(L^2(0,1))^2)} \\
&\leq Ce^{CT} \underbrace{e^{\frac{M}{T_{k+2}-T_{k+1}}} \rho_S(T_k)}_{\rho_0(T_{k+2})} \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(T_k,T_{k+1};(L^2(0,1))^2)} \\
\implies \left\| \frac{\hat{q}_{k+1}}{\rho_0} \right\|_{L^2(T_{k+1},T_{k+2})} &\leq Ce^{CT} \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(T_k,T_{k+1};(L^2(0,1))^2)} \quad \forall\ k \geq 0.
\end{aligned}$$

Step 2: Source term method

Proof of null controllability: Estimate of the control

- We now define the control q as

$$q(t) = \sum_{k \geq 0} \hat{q}_k(t) \chi_{(T_k, T_{k+1})}, \quad t \in (0, T).$$

- Note that $\|\hat{q}_0\|_{L^2(0,T_1)} \leq M e^{\frac{M}{T_1}} \|(y_0, z_0)\|_{(L^2(0,1))^2}$ and therefore

$$\left\| \frac{\hat{q}_0}{\rho_0} \right\|_{L^2(0,T_1)} \leq \frac{M}{\rho_0(T_1)} e^{\frac{M}{T_1}} \|(y_0, z_0)\|_{(L^2(0,1))^2} \leq C e^{\frac{C}{T}} \|(y_0, z_0)\|_{(L^2(0,1))^2}.$$

- Combining these estimates, we deduce

$$\left\| \frac{q}{\rho_0} \right\|_{L^2(0,T)} \leq C e^{CT + \frac{C}{T}} \left(\|(y_0, z_0)\|_{(L^2(0,1))^2} + \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(0,T;(L^2(0,1))^2)} \right).$$

- We define

$$(y, z) = (\tilde{y}, \tilde{z}) + (\hat{y}, \hat{z}).$$

- Note that $(y(0), z(0)) = a_0 = (y_0, z_0)$ and for all $k \geq 0$

$$\begin{aligned}
(y(T_{k+1}^-), z(T_{k+1}^-)) &= (\tilde{y}(T_{k+1}^-), \tilde{z}(T_{k+1}^-)) + (\hat{y}(T_{k+1}^-), \hat{z}(T_{k+1}^-)) = a_{k+1} \\
(y(T_{k+1}^+), z(T_{k+1}^+)) &= (\tilde{y}(T_{k+1}^+), \tilde{z}(T_{k+1}^+)) + (\hat{y}(T_{k+1}^+), \hat{z}(T_{k+1}^+)) = a_{k+1} \\
\implies (y, z) &\text{ is continuous at } T_k \text{ for all } k \geq 0.
\end{aligned}$$

- Moreover, we have

$$\begin{aligned}
&\left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{\mathcal{C}^0([0,T];(L^2(0,1))^2)} + \left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{L^2(0,T;(H^1(0,1))^2)} \\
&\leq C e^{CT + \frac{C}{T}} \left(\|(y_0, z_0)\|_{(L^2(0,1))^2} + \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(0,T;(L^2(0,1))^2)} \right). \quad \square
\end{aligned}$$

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Step 3: Banach fixed point argument

Introduction

- Let $T > 0, \alpha \geq 0$ and $(y_0, z_0) \in (L^2(0,1))^2$ be s.t. $\|(y_0, z_0)\|_{(L^2(0,1))^2} \leq \delta$ for some $\delta > 0$.
- We define the set

$$\mathcal{S}_\delta := \left\{ (\xi, \eta) \in L^2(0,T;(L^2(0,1))^2) \ : \ \left\| \left(\frac{\xi}{\rho_S}, \frac{\eta}{\rho_S} \right) \right\|_{L^2(0,T;(L^2(0,1))^2)} \leq \delta \right\}.$$

- Source term method implies existence of a control $q \in L^2(0,T)$ such that the solution (y,z) of (LSS) with source term $(\xi, \eta) \in \mathcal{S}_\delta$ satisfies $(y(T,\cdot), z(T,\cdot)) = (0,0)$.
- We define the map $\mathcal{F} : \mathcal{S}_\delta \to L^2(0,T;(L^2(0,1))^2)$ by

$$\mathcal{F}(\xi,\eta) := \left(f\left(y, z, \int_0^1 y, \int_0^1 z \right), g\left(y, z, \int_0^1 y, \int_0^1 z \right) \right).$$

- We prove: Existence of a $\delta > 0$ such that $\mathcal{F}$ has a unique fixed point in $\mathcal{S}_\delta$.

  $\implies$ Small time local null controllability of the nonlinear system (NL).

Lemma (Stability)

There exists $\delta > 0$ such that $\mathcal{F}(\mathcal{S}_\delta) \subset \mathcal{S}_\delta$.

Proof.

For any $(\xi, \eta) \in \mathcal{S}_\delta$, we have:

$$\left\| \frac{1}{\rho_S} \mathcal{F}(\xi, \eta) \right\|_{L^2(0,T;(L^2(0,1))^2)} \leq C \int_0^T \frac{1}{\rho_S^2(t)} \int_0^1 \left(|y(t,x)|^4 + |z(t,x)|^4 \right) dxdt$$

$$\leq C \int_0^T \frac{\rho_0^4(t)}{\rho_S^2(t)} \int_0^1 \left(\left\| \frac{y(t,\cdot)}{\rho_0(t)} \right\|^2_{L^\infty(0,1)} \left| \frac{y(t,x)}{\rho_0(t)} \right|^2 + \left\| \frac{z(t,\cdot)}{\rho_0(t)} \right\|^2_{L^\infty(0,1)} \left| \frac{z(t,x)}{\rho_0(t)} \right|^2 \right) dxdt$$

$$\leq C \left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{C^0([0,T];(L^2(0,1))^2)} \left\| \left(\frac{y}{\rho_0}, \frac{z}{\rho_0} \right) \right\|_{L^2(0,T;(H^1(0,1))^2)}$$

$$\leq C e^{CT + \frac{C}{T}} \delta^2 \quad \text{as } \|(y_0, z_0)\|_{(L^2(0,1))^2} \leq \delta, \text{ and } (\xi, \eta) \in \mathcal{S}_\delta.$$

Lemma (Contraction)

There exists $\delta > 0$ such that $\mathcal{F} : \mathcal{S}_\delta \to \mathcal{S}_\delta$ is a contraction.

Proof.

For any $(\xi_1, \eta_1), (\xi_2, \eta_2) \in \mathcal{S}_\delta$, we have:

$$\left\| \frac{1}{\rho_S} \left(\mathcal{F}(\xi_1, \eta_1) - \mathcal{F}(\xi_2, \eta_2) \right) \right\|_{L^2(0,T;(L^2(0,1))^2)} \leq C \int_0^T \frac{1}{\rho_S^2(t)} \left\| y_1(t) z_1(t) - y_2(t) z_2(t) \right\|^2_{L^2(0,1)} dt + \cdots$$

$$\leq C \int_0^T \frac{1}{\rho_S^2(t)} \left(\|y_1(t)\|^2_{L^\infty(0,1)} \|z_1(t) - z_2(t)\|^2_{L^2(0,1)} + \|y_1(t) - y_2(t)\|^2_{L^2(0,1)} \|z_2(t)\|^2_{L^\infty(0,1)} \right) dt + \cdots$$

$$\leq C \int_0^T \frac{\rho_0^4(t)}{\rho_S^2(t)} \left(\left\| \frac{y_1(t)}{\rho_0(t)} \right\|^2_{H^1(0,1)} \left\| \frac{z_1(t) - z_2(t)}{\rho_0(t)} \right\|^2_{L^2(0,1)} + \left\| \frac{y_1(t) - y_2(t)}{\rho_0(t)} \right\|^2_{L^2(0,1)} \left\| \frac{z_2(t)}{\rho_0(t)} \right\|^2_{H^1(0,1)} \right) dt + \cdots$$

$$\leq C \left\| \left(\frac{y_1}{\rho_0}, \frac{z_1}{\rho_0} \right) - \left(\frac{y_2}{\rho_0}, \frac{z_2}{\rho_0} \right) \right\|^2_{C^0([0,T;(L^2(0,1))^2])} \times \left\| \left(\frac{y_1}{\rho_0}, \frac{z_1}{\rho_0} \right) \right\|^2_{L^2(0,T;(H^1(0,1))^2)} + \cdots$$

Small-time local null controllability

Theorem

The nonlinear system (NL) is small-time locally null controllable in $(L^2(0,1))^2$ *around the equilibrium point* $(0,0)$.

Future directions

- Let $T > 0, \alpha \geq 0$ be given. We consider the following system:

$$(NL) \quad \begin{cases} \partial_t y - \partial_x(\gamma_1 \partial_x y) = f(y, z, \int_0^1 y, \int_0^1 z), & \text{in } (0,T)\times(0,1), \\ \partial_t z - \partial_x(\gamma_2 \partial_x z) = g(y, z, \int_0^1 y, \int_0^1 z), & \text{in } (0,T)\times(0,1), \\ \partial_x y(t,0) = q(t), \quad \partial_x z(t,0) = 0, & \text{for } t \in (0,T), \\ \partial_x y(t,1) = \partial_x z(t,1), & \text{for } t \in (0,T), \\ y(t,1) + z(t,1) + \alpha \partial_x y(t,1) = 0, & \text{for } t \in (0,T), \\ y(0,x) = y_0(x), \quad z(0,x) = z_0(x), & \text{in } (0,1). \end{cases}$$

- $\gamma_1, \gamma_2 \in L^\infty((0,T)\times(0,1))$.
- Prove small-time local null controllability in $(L^2(0,1))^2$ around the equilibrium $(0,0)$.
- Similar problem in higher dimension (using Carleman estimate).

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Thank you for your attention.